TrustGate - Hardware Security Monitor

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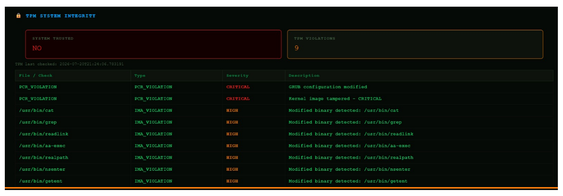
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INTRODUCTION

In 2021, an attacker remotely accessed a Florida water treatment plant and tried to raise sodium hydroxide to 100x its safe level. The breach wasn't caught by any existing security tool because those tools watch the network, not the plant. They tell you a register changed; they don't tell you that a pipe is about to fail, or that a chemical is now lethal.

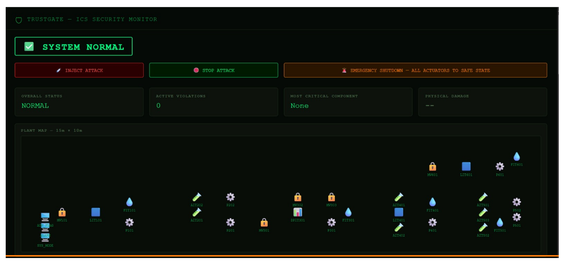
TrustGate is a hardware-rooted monitoring system for water-treatment plants that converts cyber and sensor telemetry into actionable physical insight. It detects anomalous activity, estimates its confidence, classifies the likely attack type, and localizes the probable source within the plant. Results are communicated through both a spatial visualization and a ranked textual explanation, enabling operators to identify not only that an attack is occurring, but also its likely type, severity, and physical location.

FEATURES



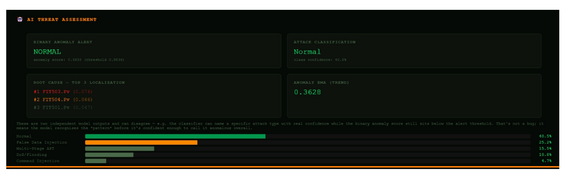
Hardware Trust and Watchdog

Hardware Trust TPM attestation verifies boot-time PCR measurements and runtime integrity. A hardware watchdog supports automatic recovery when the monitoring process becomes unresponsive.



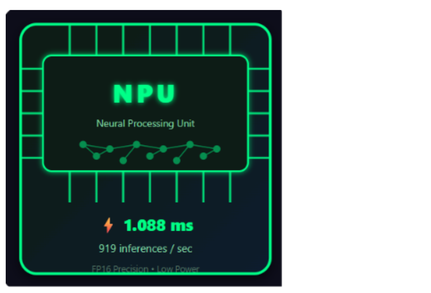
Physical–Cyber Context

The digital twin connects sensor behaviour and network activity to real plant assets. Events are mapped to their physical location rather than reported only as register-level anomalies.



Multi-Head AI Engine

A dual-stream sensor–network BiLSTM with cross-attention produces three outputs in one forward pass: binary anomaly detection, six-class attack classification, and ranked root-cause localization.



NPU-Accelerated Inference

OpenVINO deploys the AI pipeline on the Intel NPU instead of the CPU, supporting efficient edge inference without cloud dependency.

RESULTS & ANALYSIS

TrustGate is evaluated on three axes: known-attack detection, zero-shot generalization to unseen components (MV302, MV303, AIT402 — never attacked in training), and what the gap between the two reveals about the role of plant-specific context in ICS security.

Known-Attack Detection (Validation Set)

Method	AUC	F1	FAR
Isolation Forest	0.1057	0.1412	99.34%
One-Class SVM+PCA	0.2827	0.1609	90.22%
Autoencoder	0.2446	0.1717	83.17%
Simple BiLSTM	0.7776	0.3113	22.30%
TrustGate v3	**0.9620**	**0.7389**	**2.97%**

Classical detectors (AUC 0.10–0.28, FAR >80%) score timesteps independently and miss the multi-second temporal escalation of ICS attacks. The unimodal BiLSTM (0.7776) shows sequence modeling helps, but the remaining 0.18 AUC / 7× FAR gain comes specifically from dual-stream fusion + cross-attention (below).

Ablation Study

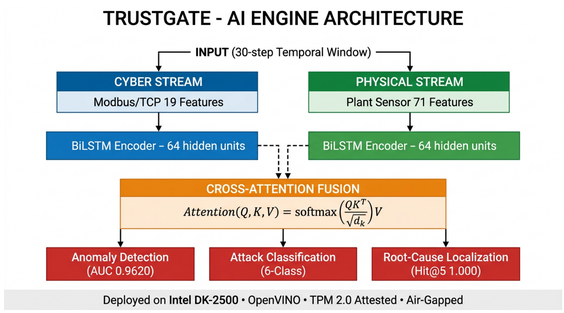
Config	AUC	Δ
Full (dual+attn.)	**0.9620**	—
Sensor only	0.0943	−0.8677
Network only	0.8591	−0.1029
Concat, no attn.	0.7422	−0.2198

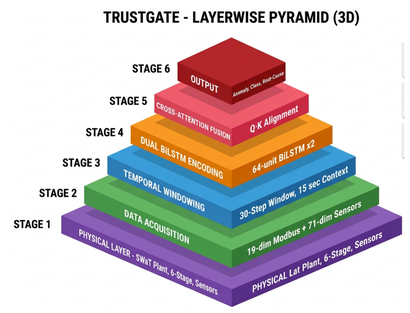
Root-Cause Localization

Metric	Known	Zero-Shot
Hit@5	0.4073	**1.0000**
MRR	0.3227	**1.0000**

Sensor-only AUC (0.0943, worse than random) proves physical telemetry alone cannot detect masked attacks disguised as normal operational transitions — network behavior carries most of the signal (0.8591 network-only). **Concatenation without attention (0.7422)** shows simply combining streams isn't enough: naive fusion destroys the temporal cyber–physical correlation. Only cross-attention, letting each stream query the other ("which sensor state correlates with this command, at this lag?"), recovers the full 0.9620.

SYSTEM ARCHITECTURE





Why BiLSTM?
To effectively capture both past temporal dependencies and forward context across dual streams of Modbus network vectors (19-dim) and physical sensor states (71-dim), enabling rapid detection of subtle anomalies.

Is the system itself protected?
To secure this model against tampering and operational failures, the execution pipeline is protected using TPM 2.0 boot attestation, continuous runtime Integrity Measurement Architecture (IMA) verification, and an air-gapped hardware watchdog that automatically restarts hanging processes.

FUTURE WORK

- **Full Component Coverage:** Expand training to all 22 monitored components, closing the zero-shot AUC gap entirely.
- **Physics-Informed Loss:** Mass-balance regularization (PINN) to catch physically-impossible sensor spoofing.
- **Batched NPU Inference:** Batch size 16–32 to amortize dispatch overhead for 24/7 energy efficiency.
- **Federated Multi-Plant Learning:** Share attack knowledge across plant deployments while preserving plant-specific context.